

ProjectSummary

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Variability of Frequency/Positivity

Modifying the Kernel XY Values

Parameters varied for Simulations: The maximum and minimum values for the kernel window were changed, the final results were displayed with resulting range of x and y values (the range of x values used for a simulation were equal to the range of y values used in that same simulation so only the one range/distance is reported).

```
maxs <- seq(9, 6.5, -0.5)
mins <- seq(1, 3.5, 0.5)
kdis <- maxs - mins; kdis
```

```
## [1] 8 7 6 5 4 3
```

```
objs <- readRDS('kernXY_OBJS.rds')
```

Below are the parameters that were used for the first simulation. The ymax, ymin, xmax, and xmin values for the tissue were changed for subsequent simulations while all other parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

```
## Holes has not yet been simulated
```

```
pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
names(pv) <- names(param)
pv
```

```
## $Tissue
```

```
##      k  xmin  xmax  ymin  ymax sdmin sdmax
## 10.0   0.0  10.0   0.0  10.0   0.5   2.0
##
```

```
## $Holes
```

```
##      xmin      xmax      ymin      ymax      sdmin      sdmax hole_prob1
##      0.00     10.00      0.00     10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
```

```
## $Cells
```

```
##      Cell Type 1.k  Cell Type 1.xmin  Cell Type 1.xmax  Cell Type 1.ymin
##              1.0              1.0              9.0              1.0
##      Cell Type 1.ymax  Cell Type 1.sdmin  Cell Type 1.sdmax  Cell Type 1.probs1
```

```
##          9.0          0.5          2.0          0.0
## Cell Type 1.probs2
##          1.0
```

Calculations of frequencies and boxplots.

```
ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2' ) /
      sum(ca[[i]][[j]][1] == 'Tissue 2')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2' )
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

cn <- list(paste0('KernelDistances=', kdis))[[1]]
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames

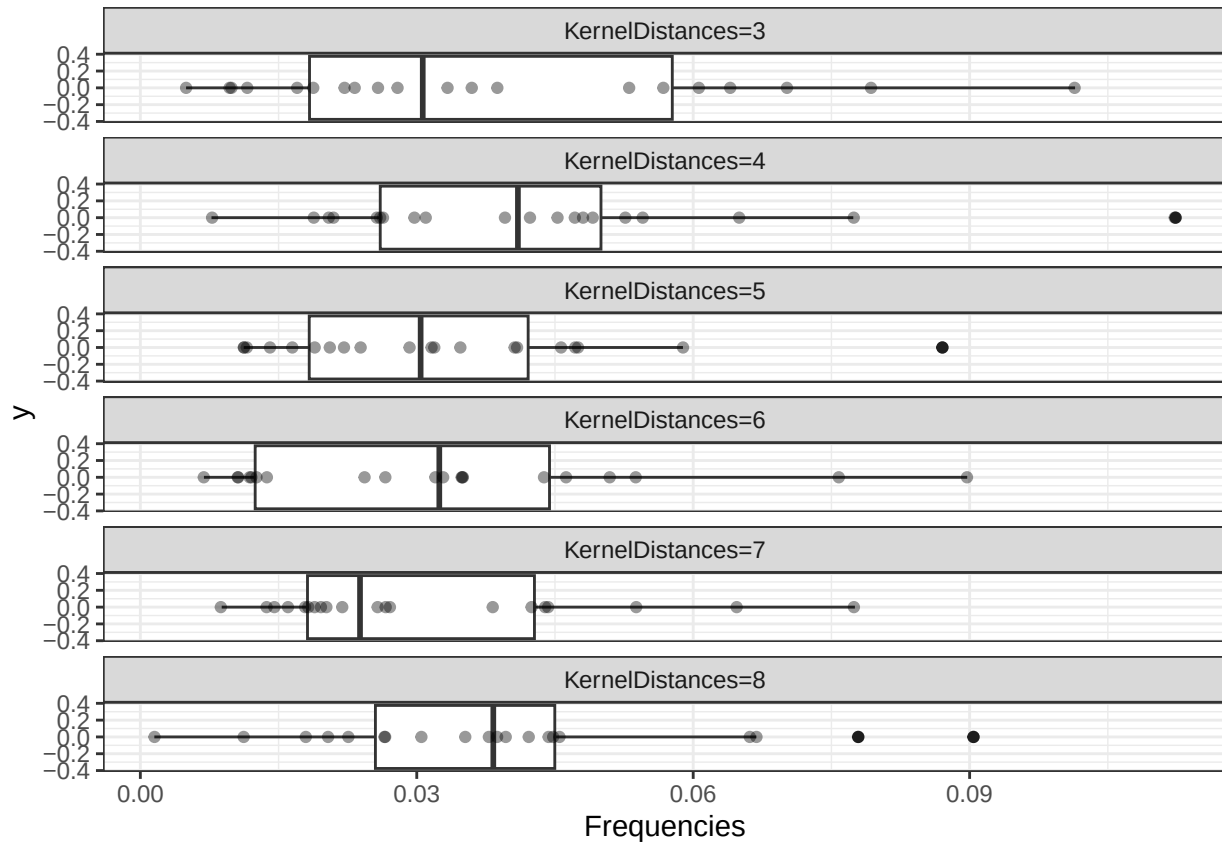
d.f1.long <- pivot_longer(d.f1,
                          cols = all_of(cn),
                          names_to = "kdis",
                          values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
                          cols = all_of(cn),
```

```
names_to = "kdis",
values_to = "Positives")
```

```
# create boxplots
```

```
freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ kdis, ncol = 1) +
  theme_bw();freq.box
```



```
pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ kdis, ncol = 1) +
  theme_bw();pos.box
```

